

Liver Cirrhosis Clinical Analysis

Patient Overview

Demographics · Population snapshot · 25,000 patients

Total Patients
25K

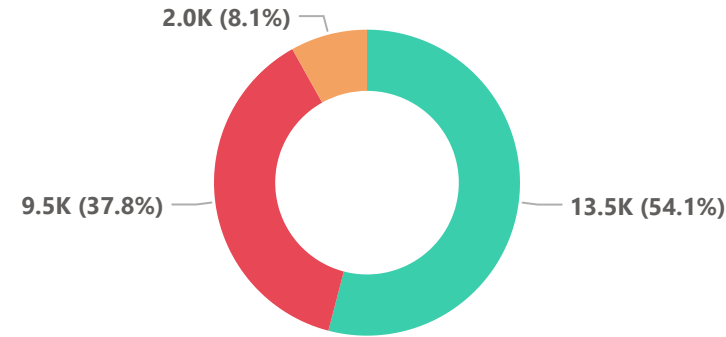
Mortality Rate
37.8%

Censored (Alive)%
54.1%

Avg Follow-up Days
1,887

Patient Status Breakdown

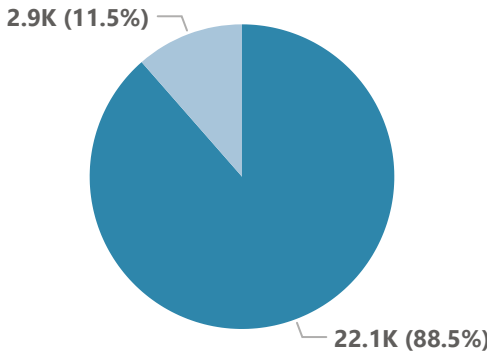
54.1% of patients were censored (alive at study end) · 37.8% deceased · 8.1% received liver transpl...



Status ● Censored (Alive) ● Deceased ● Liver Transplant

Patient Gender Distribution

88.5% female — consistent with PBC predominantly affecting women at a 9:1 ratio



Sex ● F ● M

Treatment

Drug

Placebo

Gender

F

M

Disease Stage

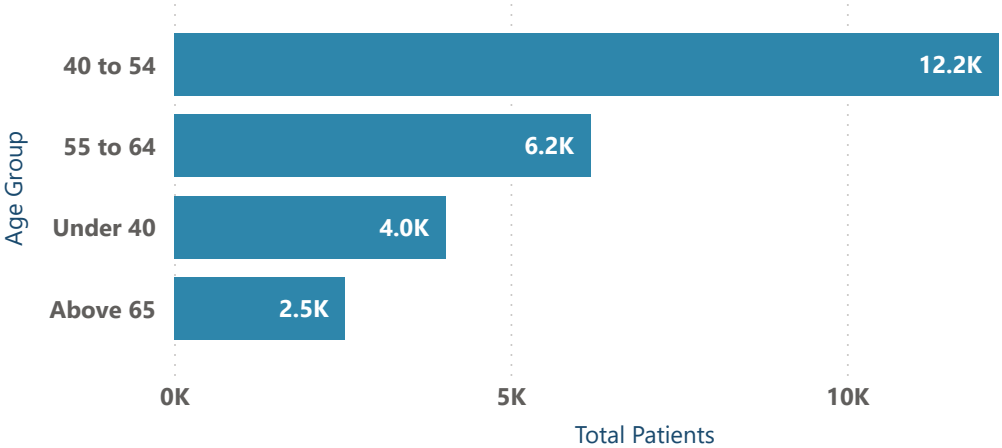
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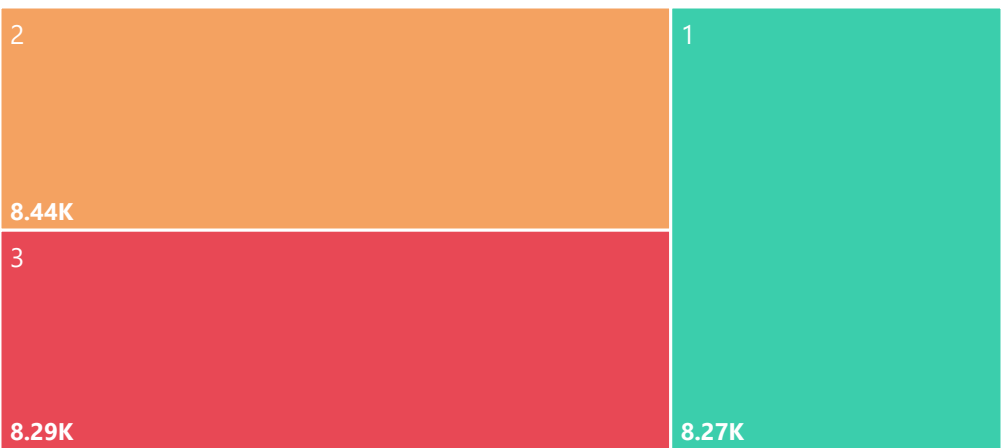
Patient Age Group Distribution

49.2% of patients aged 40–54 · mean age 50.3 years · range 26–78



Disease Stage Distribution

Stage ● 2 ● 3 ● 1



This dataset of 25,000 liver cirrhosis patients shows a near equal distribution across disease stages, with an 88.5% female majority, consistent with PBC clinical epidemiology. 37.8% of patients were deceased at study end, underscoring the

Symptom Prevalence

Clinical signs · How common · Progression by stage

Treatment

Drug

Placebo

Ascites Rate

32.8%

Hepatomegaly Rate

39.0%

Spiders Rate

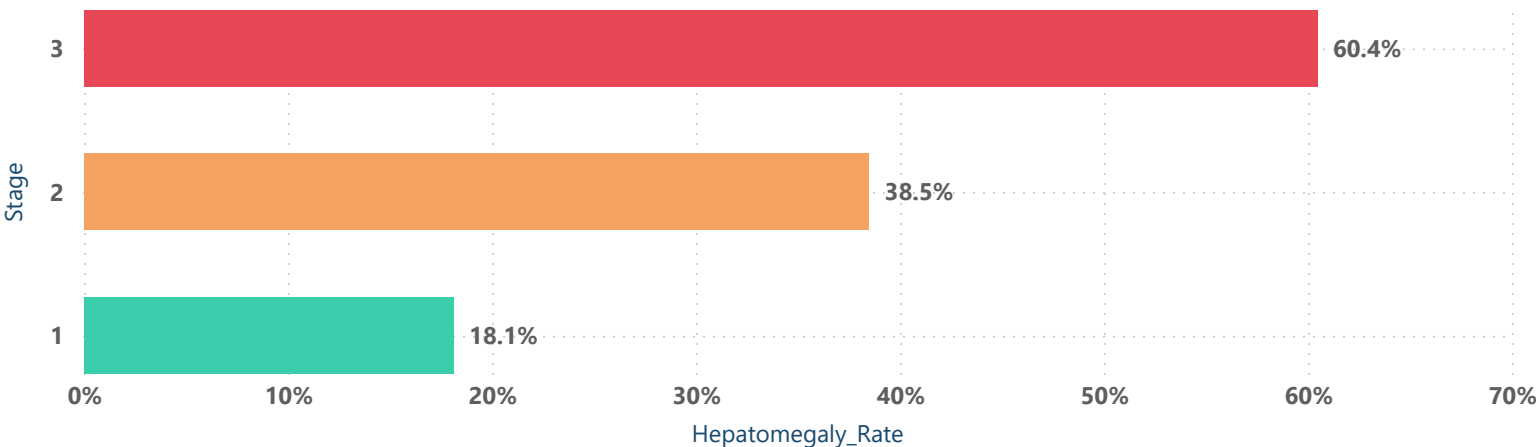
45.5%

Edema Rate

19.1%

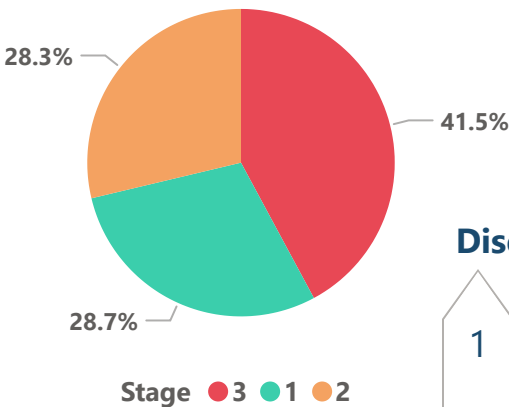
Hepatomegaly Prevalence by Disease Stage

Present in 60.4% of Stage 3 patients vs 18.1% at Stage 1 — strongest clinical progression marker in the dataset

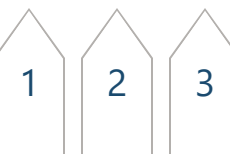


Ascites Prevalence by Disease Stage

Stable across Stages 1–2 then rises sharply at Stage 3 — consistent with late...

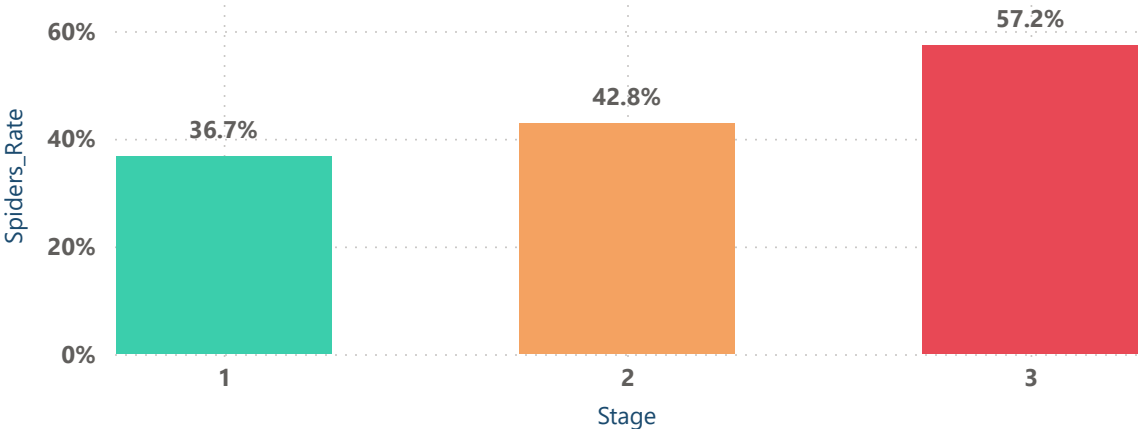


Disease Stage



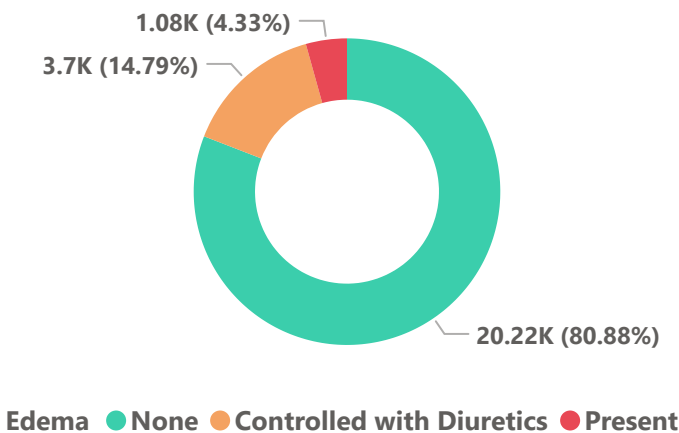
Spider Angiomata Prevalence by Disease Stage

Rises steadily from 36.7% at Stage 1 to 57.2% at Stage 3 — consistent with progressive portal hy...



Edema Severity Distribution

19.1% of patients show some form of edema · severe edema present in 4...



Hepatomegaly is the strongest disease progression marker in this dataset, present in only 18.1% of Stage 1 patients but rising to 60.4% at Stage 3. Spider and Ascites also increase with severity, while edema though present in only 19.1% overall, carries critical survival implications explored in the following impact Dashboard.

Symptom Impact

Mortality by symptom · Edema severity · Combinations

Mortality Rate

37.8%

Mortality No Edema

32.8%

Mortality Edema Controlled

49.9%

Mortality Severe Edema

90.8%

Gender



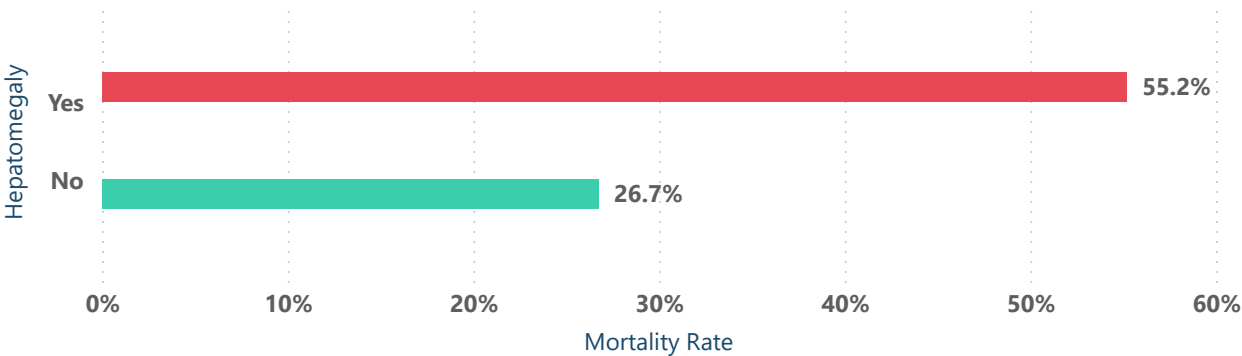
Disease Stage



Mortality Rate by Hepatomegaly Presence

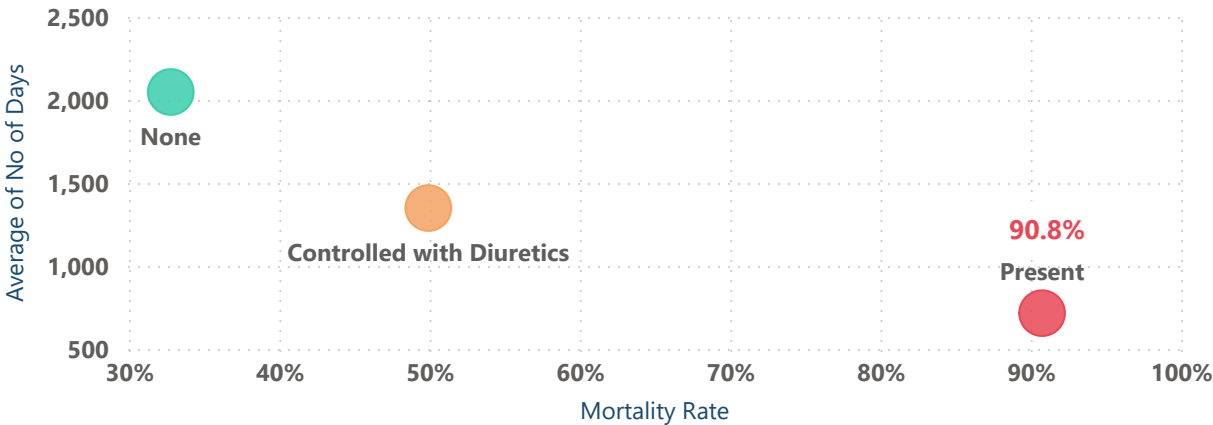
55.2% mortality when present vs 26.7% when absent — more than double the risk

Hepatomegaly ● Yes ● No



Edema Severity by Mortality Rate and Avg Follow up days

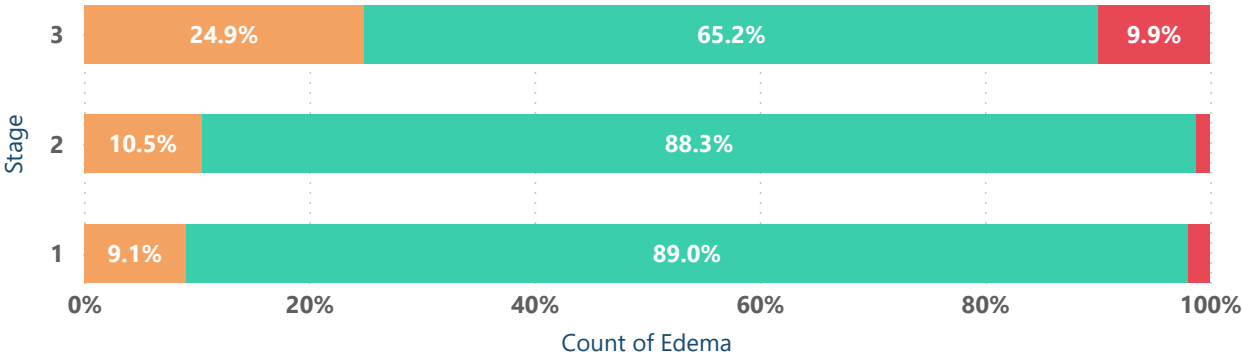
Avg follow-up just 716 days for severe edema patients vs 2,048 for no edema · carries 90.8% mortality r...



Edema Severity Distribution by Disease Stage

Severe edema rises fivefold from 1.9% at Stage 1 to 9.9% at Stage 3

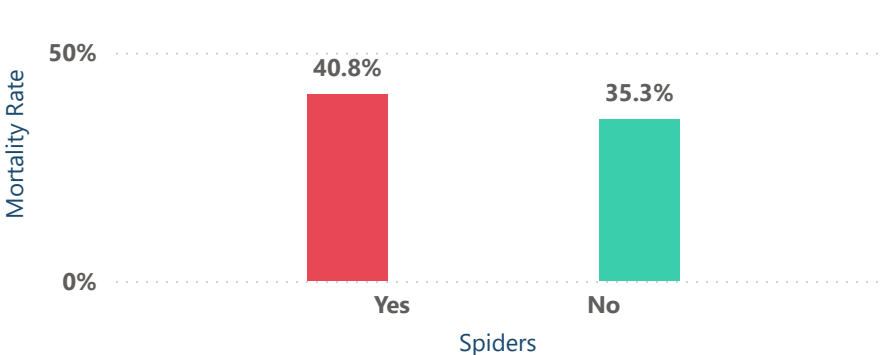
Edema ● Controlled with Diuretics ● None ● Present



Mortality Rate by Spiders Presence

40.8% mortality when present vs 35.3% when absent

Spiders ● Yes ● No



Hepatomegaly presence more than doubles mortality risk, 55.2% when present vs 26.7% when absent. Severe edema is the most critical single indicator carrying a 90.8% mortality rate and an average follow up of just 716 days.

Biomarker Averages

Lab values by stage · Clinical reference ranges

Avg Bilirubin (mg/dL)
3.40

Avg Albumin (g/dL)
3.49

Avg Platelets (×10³/μL)
256.01

Avg Copper (μg/day)
100.18

Gender

F

M

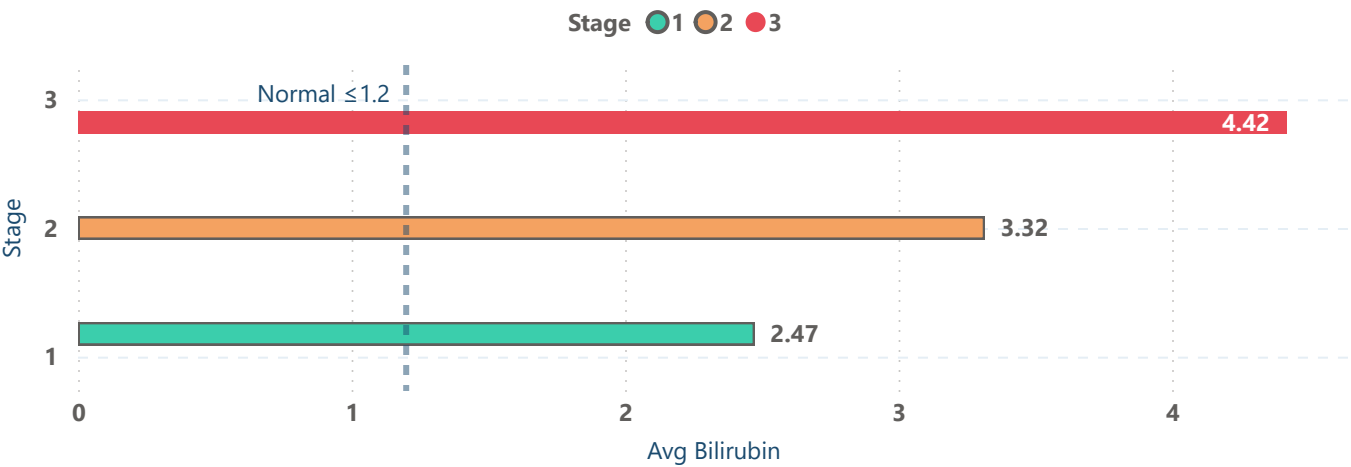
Treatment

Drug

Placebo

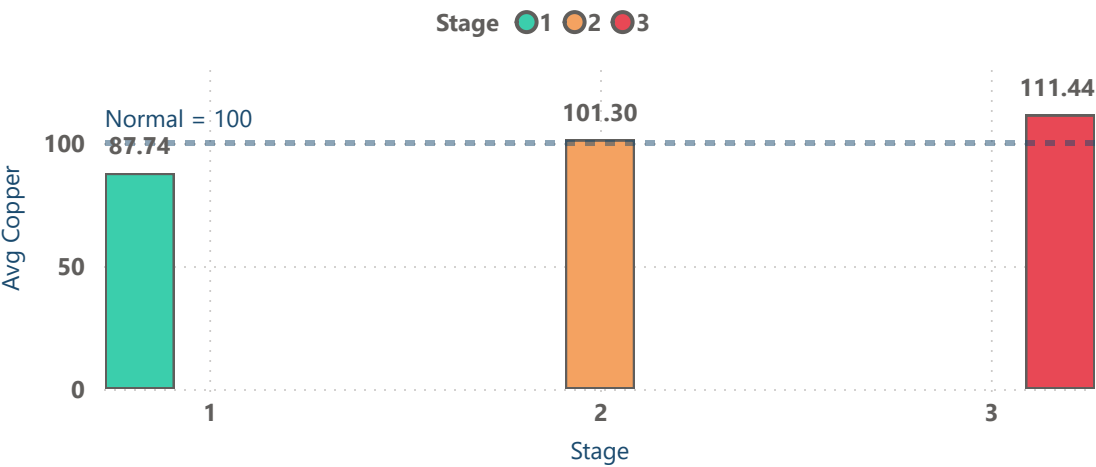
Average Bilirubin by Disease Stage

All stages above normal limit (≤1.2 mg/dL) · Stage 3 is 79% higher than Stage 1



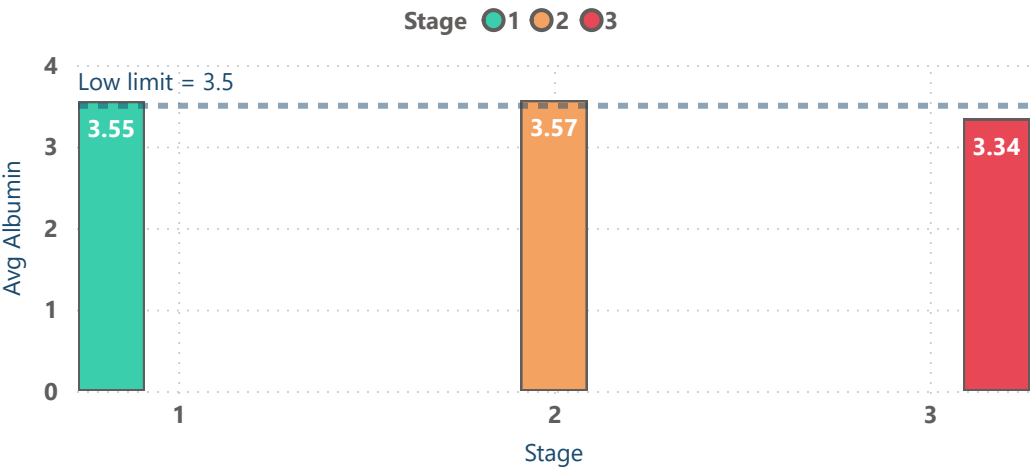
Average Copper Level by Disease Stage

Copper exceeds normal limit (≤100 μg/day) from Stage 2 onwards — a marker of progressive...



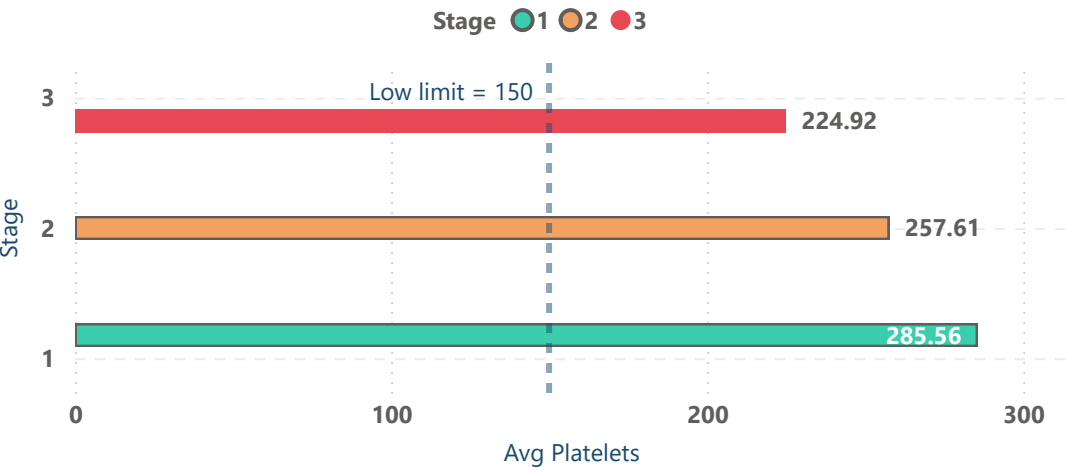
Average Albumin by Disease Stage

Stage 3 falls below the 3.5 g/dL threshold — indicating protein synthesis severity



Average Platelet Count by Disease Stage

Platelets decline 21% from Stage 1 to Stage 3 — consistent with portal hypertension progress...



Bilirubin rises 79% from Stage 1 to Stage 3, while Platelets drop 21%, both significantly outside normal clinical ranges at Stage 3. Copper exceeds the normal threshold from Stage 2, indicating progressive biliary obstruction preceding end stage disease.

Biomarker Severity

Scatter analysis · Composite Severity Score

Avg Severity — Stage 1

4.45

Avg Severity — Stage 2

5.25

Avg Severity — Stage 3

7.25

Gender

F

M

Disease Stage

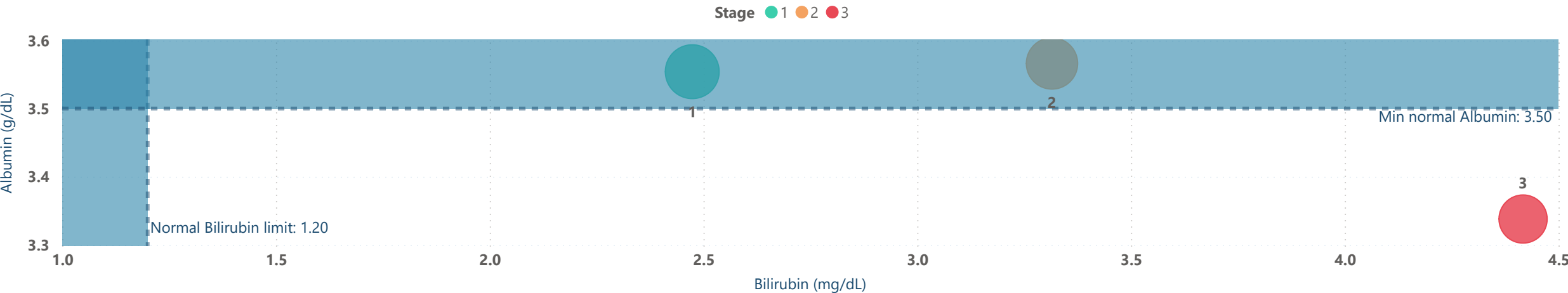
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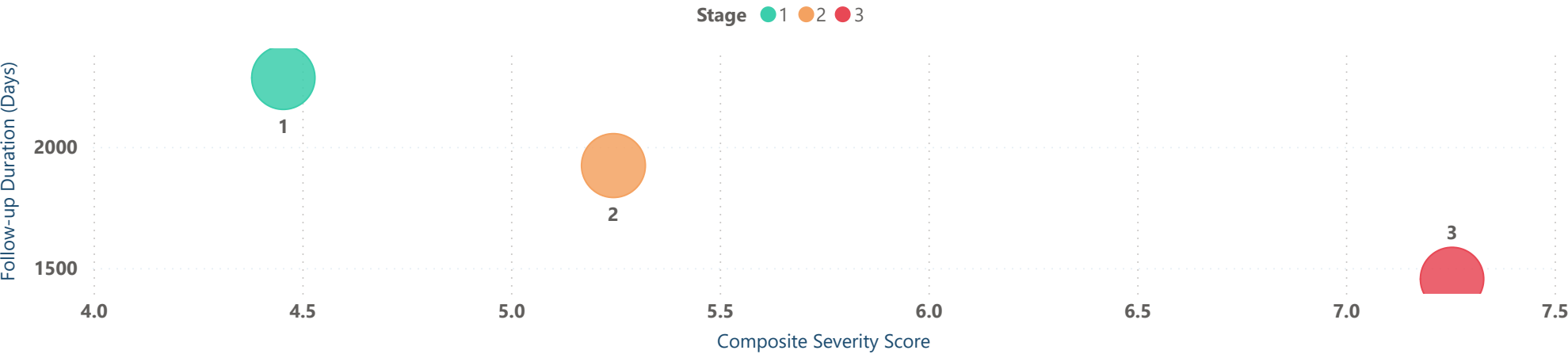
Bilirubin vs Albumin — Platelet Count & Disease Stage

Dot position = liver function · Dot size = platelet count · Dot color = disease stage



Severity Score vs Survival Outcome by Stage

Composite index (Bilirubin + Albumin deficit + Prothrombin excess) plotted against follow-up days — stratified by disease stage



The Composite Severity Score derived from Bilirubin, Albumin deficit and Prothrombin excess rises from 4.45 at Stage 1 to 7.25 at Stage 3. This 63% increase in severity corresponds to a 36% reduction in average follow up days, validating the score as a clinically meaningful composite index of disease progression.

Survival Outcome

Follow-up duration · Mortality patterns · Age impact

Overall Avg days

1,887

Stage 1 Avg days

2,285

Stage 2 Avg days

1,923

Stage 3 Avg days

1,455

Disease Stage

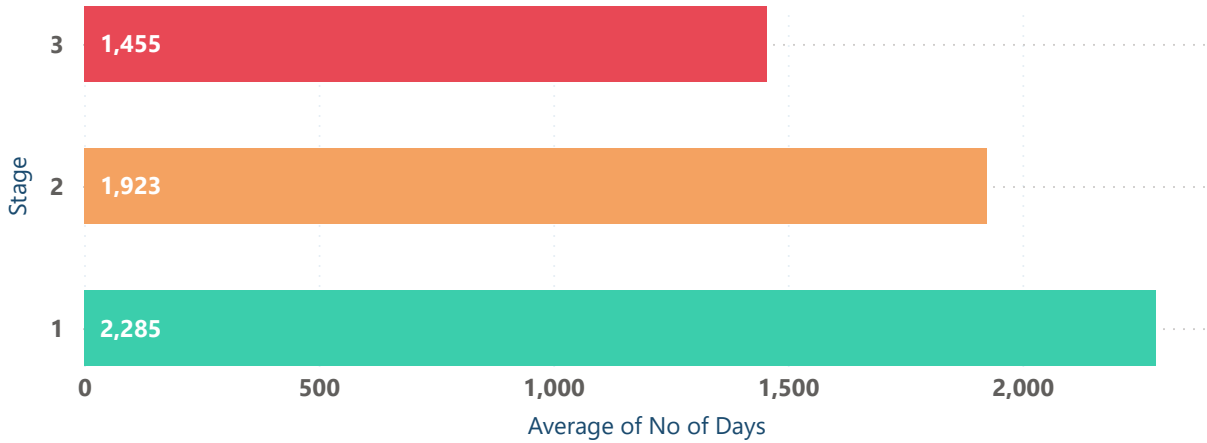


Gender



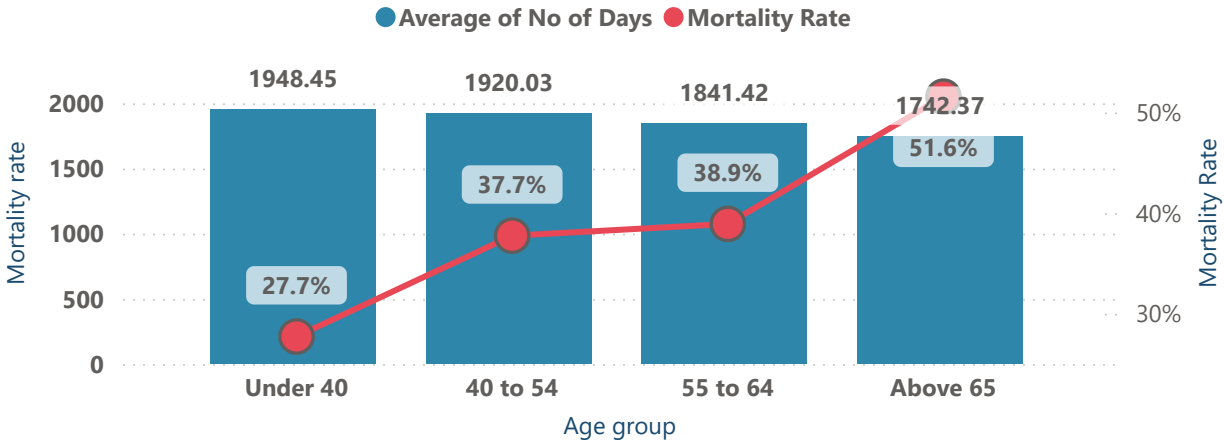
Average Follow-up Duration by Disease Stage

Stage 3 patients average 830 fewer days than Stage 1 — a 36.3% reduction in survival time



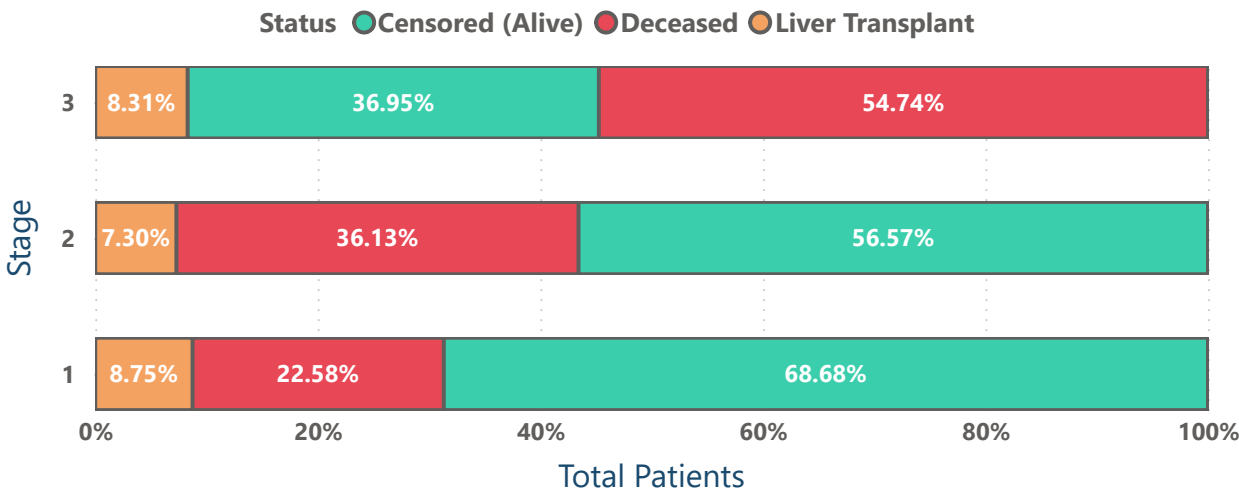
Survival Duration & Mortality Rate by Patient Age Group

Mortality nearly doubles from 27.7% (under 40) to 51.6% (65+) — confirming age as a key survival factor



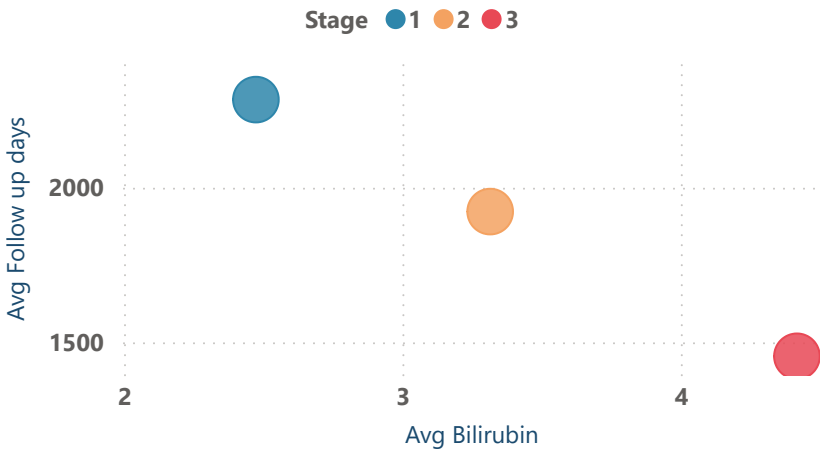
Status breakdown by stage

Mortality rises from 22.6% at Stage 1 to 54.7% at Stage 3 — more than doubling across disease progression



Bilirubin Level vs Survival Duration by Disease Stage

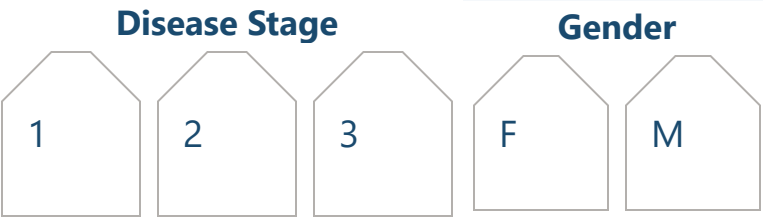
Rising bilirubin is strongly associated with reduced follow-up duration...



Stage 3 patients survive on average 830 fewer days than Stage 1 patients, a 36.3% reduction. Mortality more than doubles from 22.6% to 54.7% across disease stages. Age compounds this risk, with patients aged above 65 showing a 51.6% mortality rate versus 27.7% for those under 40.

Treatment Effectiveness

D-penicillamine vs Placebo · Stage-by-stage comparison



Drug Avg Days
2,017

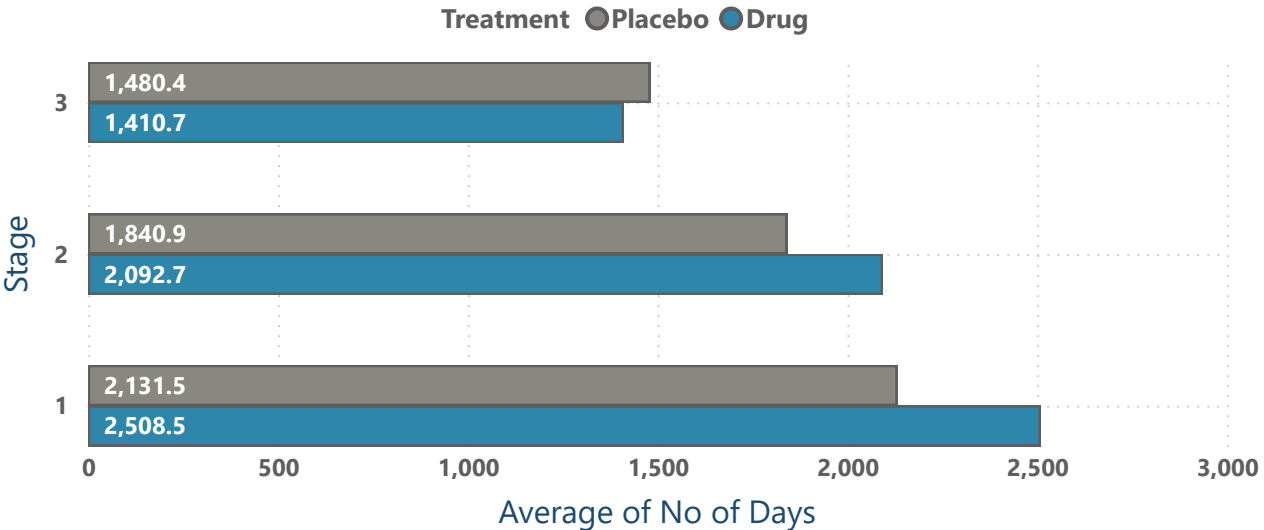
Placebo Avg Days
1,812

Drug Mortality
37.6%

Placebo Mortality
37.9%

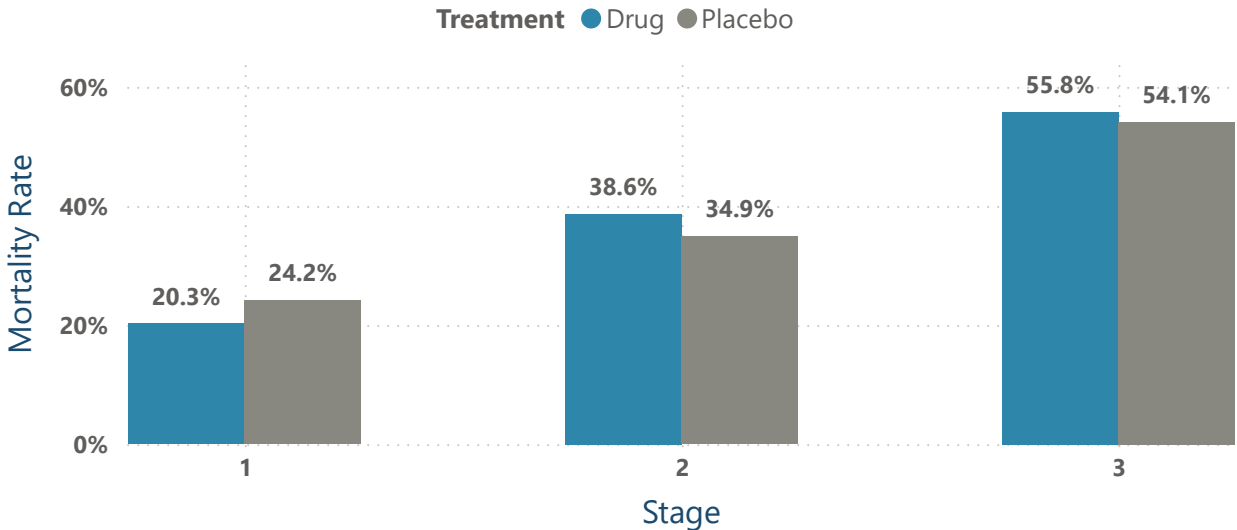
Average Survival Duration by Treatment Group & Disease Stage

Drug shows 377 more follow-up days than Placebo at Stage 1 — benefit diminishes at Stage 3

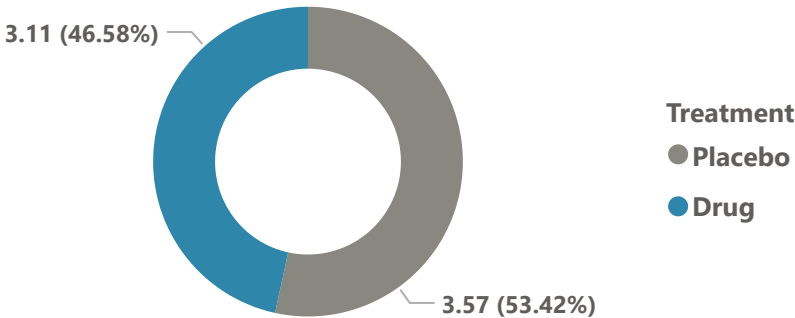


Mortality Rate by Treatment Group & Disease Stage

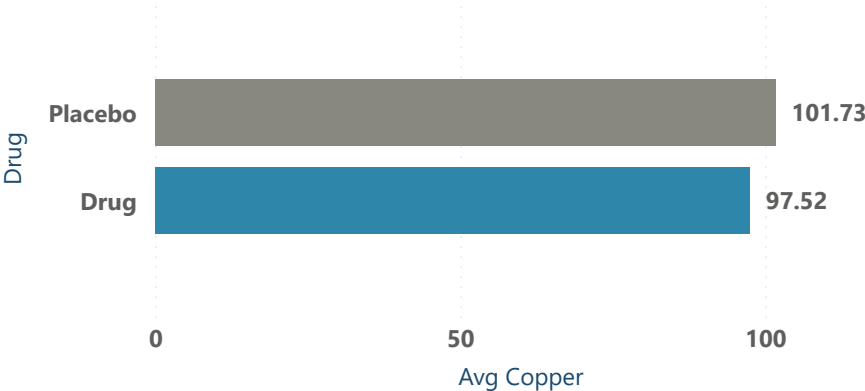
D-penicillamine shows lowest mortality at Stage 1 — benefit is not sustained in advanced disease



Avg Bilirubin by Treatment



Avg Copper by Treatment



Drug helps most at Stage 1 demonstrating a 377 day survival and 4% lower mortality rate, but the advantage disappears at Stages 2 and 3.

Drug group shows lower Bilirubin and lower Copper suggesting partial hepatic response.

⚠ This analysis is observational. Correlation between treatment and survival does not imply causation. Data source: Mayo Clinic PBC Trial 1974–1984. For academic purposes only.

Liver Cirrhosis Clinical Analysis Key Insights

Mayo Clinic PBC Trial, 1974 to 1984, 25,000 patients, 7 interactive dashboards, For academic and research purposes only

Total Patients

25K

Overall Mortality

37.8%

Avg Follow-up Days

1,887

Female Patients

88.5%

01 · Patient Demographics

88.5% Female

9:1 ratio — consistent with PBC clinical epidemiology
Age range 26–78 · mean age 50.3 years · Stages split near-equally at ≈33% each — providing an unbiased foundation for cross-stage comparison.

02 · Symptom Progression

60.4% Hepatomegaly
at Stage 3

Up from 18.1% at Stage 1 — a 3.3× rise

Strongest progression marker in the dataset. Severe edema carries 90.8% mortality and just 716 average follow-up days — versus 2,048 days for patients with no edema.

03 · Survival Outcomes

–830 Days

Stage 3 vs Stage 1 — a 36.3% reduction in survival time

Mortality more than doubles: 22.6% at Stage 1 rising to 54.7% at Stage 3. Age compounds risk — patients aged 65+ reach 51.6% mortality versus 27.7% for those under 40.

04 · Biomarker Analysis

+79% Bilirubin

2.47 → 4.42 mg/dL across disease stages

All biomarkers breach clinical norms at Stage 3. Platelets drop 21%, Copper exceeds its limit from Stage 2, and Albumin falls below the 3.5 g/dL threshold — all simultaneously.

05 · Composite Severity Score

4.45 → 7.25 Severity
Score

+62.9% rise from Stage 1 to Stage 3

MELD-inspired composite index combining Bilirubin, Albumin deficit and Prothrombin excess. Higher score correlates directly with fewer survival days — validated across all 3 stages.

06 · Treatment Effectiveness

+377 Days at Stage 1

Drug advantage exists early — disappears at Stage 3

D-penicillamine patients average 2,508 days vs 2,131 for Placebo at Stage 1. Overall mortality near-identical (37.6% vs 37.9%) — confirming early treatment is the critical window.